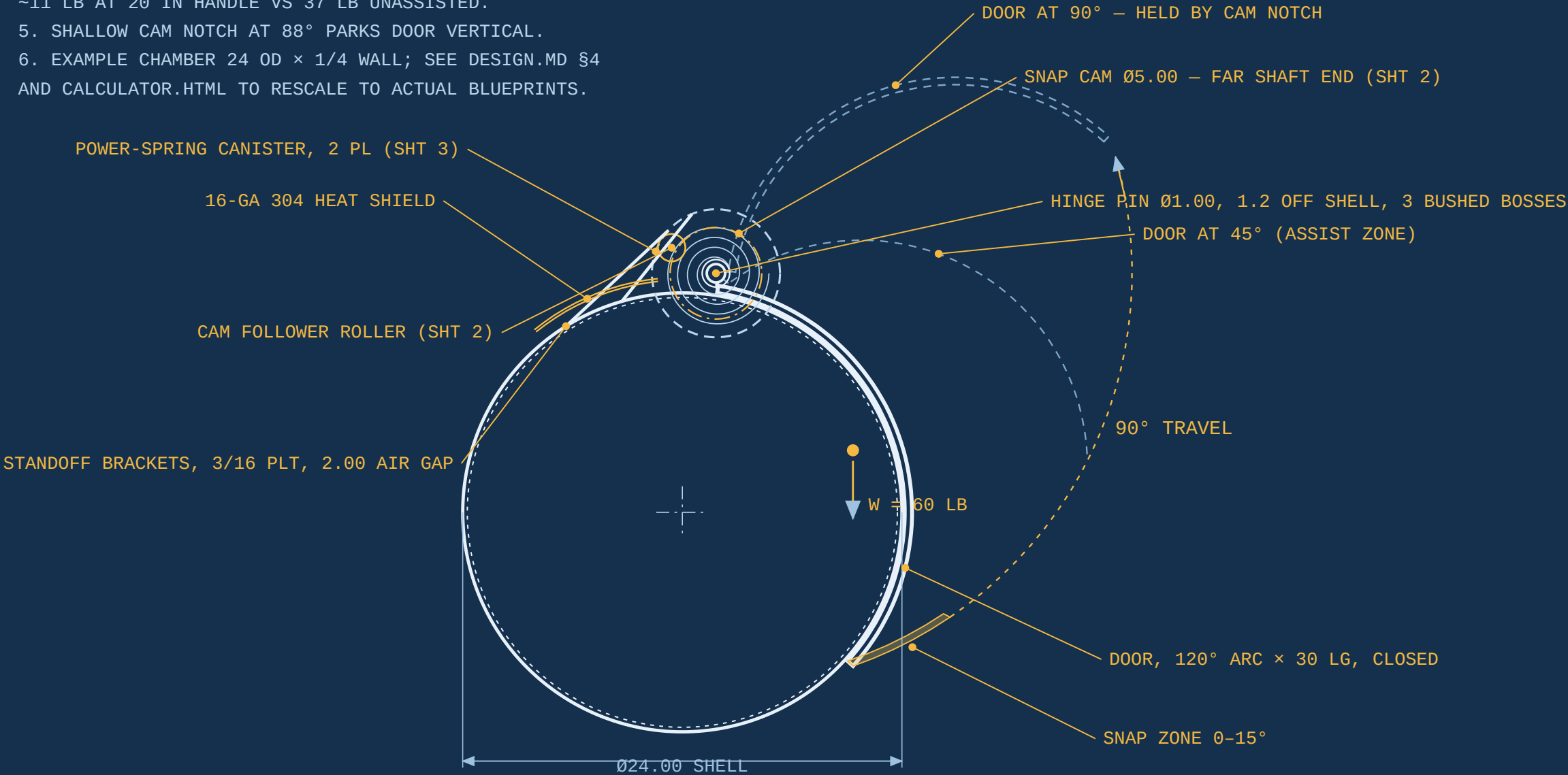


NOTES:

- 1. DOOR + HINGE SHAFT ROTATE TOGETHER; SHAFT RIDES IN 3 GRAPHITE-BRONZE BUSHED WELD BOSSES ON SHELL.
- 2. ALL SPRING HARDWARE OUTBOARD, 2.00 AIR GAP BEHIND POLISHED SS SHIELD. NO SPRING TOUCHES HOT STEEL.
- 3. CAM DETENT PULLS DOOR SHUT LAST 15° (~105 IN-LB SEAL).
- 4. SPRINGS CARRY ~73% OF DOOR WEIGHT: PEAK HAND EFFORT ~11 LB AT 20 IN HANDLE VS 37 LB UNASSISTED.
- 5. SHALLOW CAM NOTCH AT 88° PARKS DOOR VERTICAL.
- 6. EXAMPLE CHAMBER 24 OD × 1/4 WALL; SEE DESIGN.MD §4 AND CALCULATOR.HTML TO RESCALE TO ACTUAL BLUEPRINTS.



OFFSET SMOKER — DOOR SPRING-HINGE ASSEMBLY

ASSEMBLY — END VIEW, DOOR 0°/45°/90°

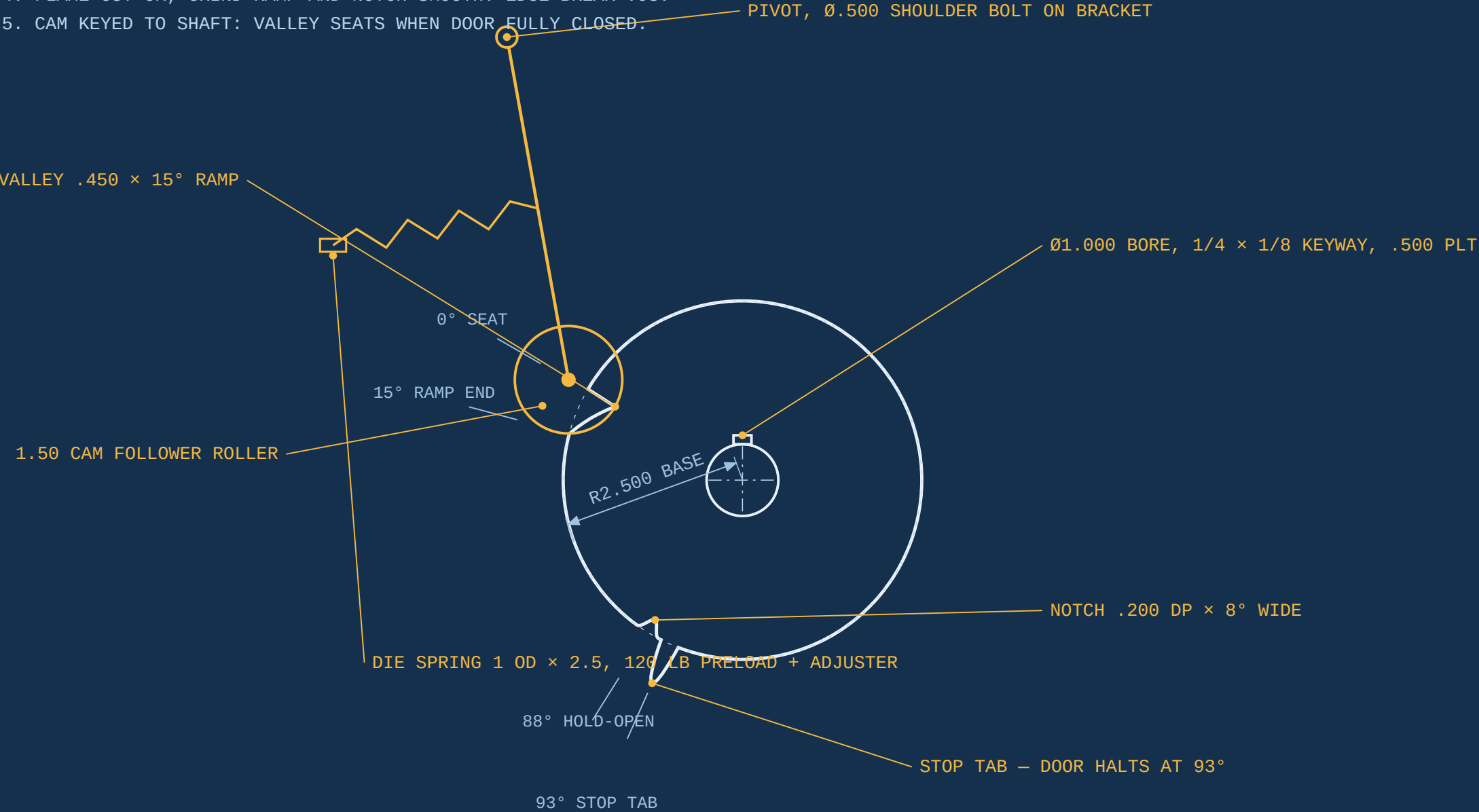
SHT 1/4 · 14 PX/IN

UNITS: INCHES · REV A — PROPOSAL

2026-07-04

NOTES:

- 1. VALLEY RAMP: .450 RISE OVER 15°, LINEAR, BLEND R.12 ENDS.
- SEAT DETENT TORQUE = 120 LB × .450/.262 RAD ≈ 206 IN-LB.
- 2. HOLD-OPEN NOTCH TORQUE ≈ 172 IN-LB.
- 3. DWELL 15°-84° IS CONSTANT RADIUS — ZERO CAM TORQUE IN ASSIST ZONE; DOOR FEEL THERE IS PURE RIBBON SPRING.
- 4. FLAME-CUT OK; GRIND RAMP AND NOTCH SMOOTH. EDGE BREAK .03.
- 5. CAM KEYED TO SHAFT: VALLEY SEATS WHEN DOOR FULLY CLOSED.



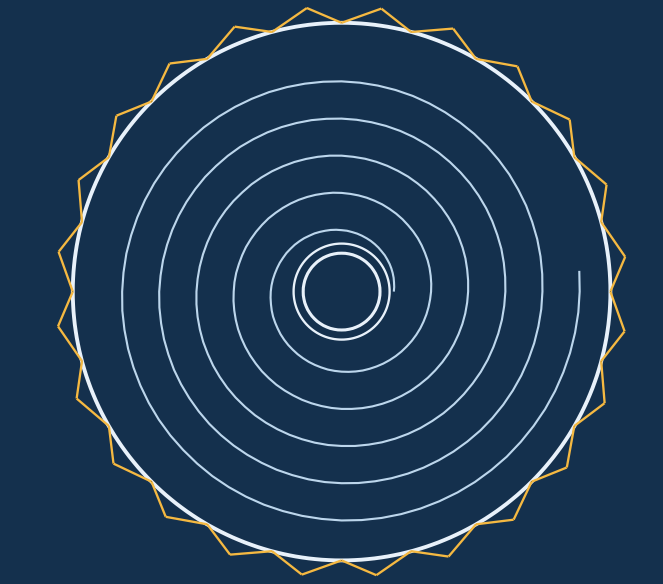
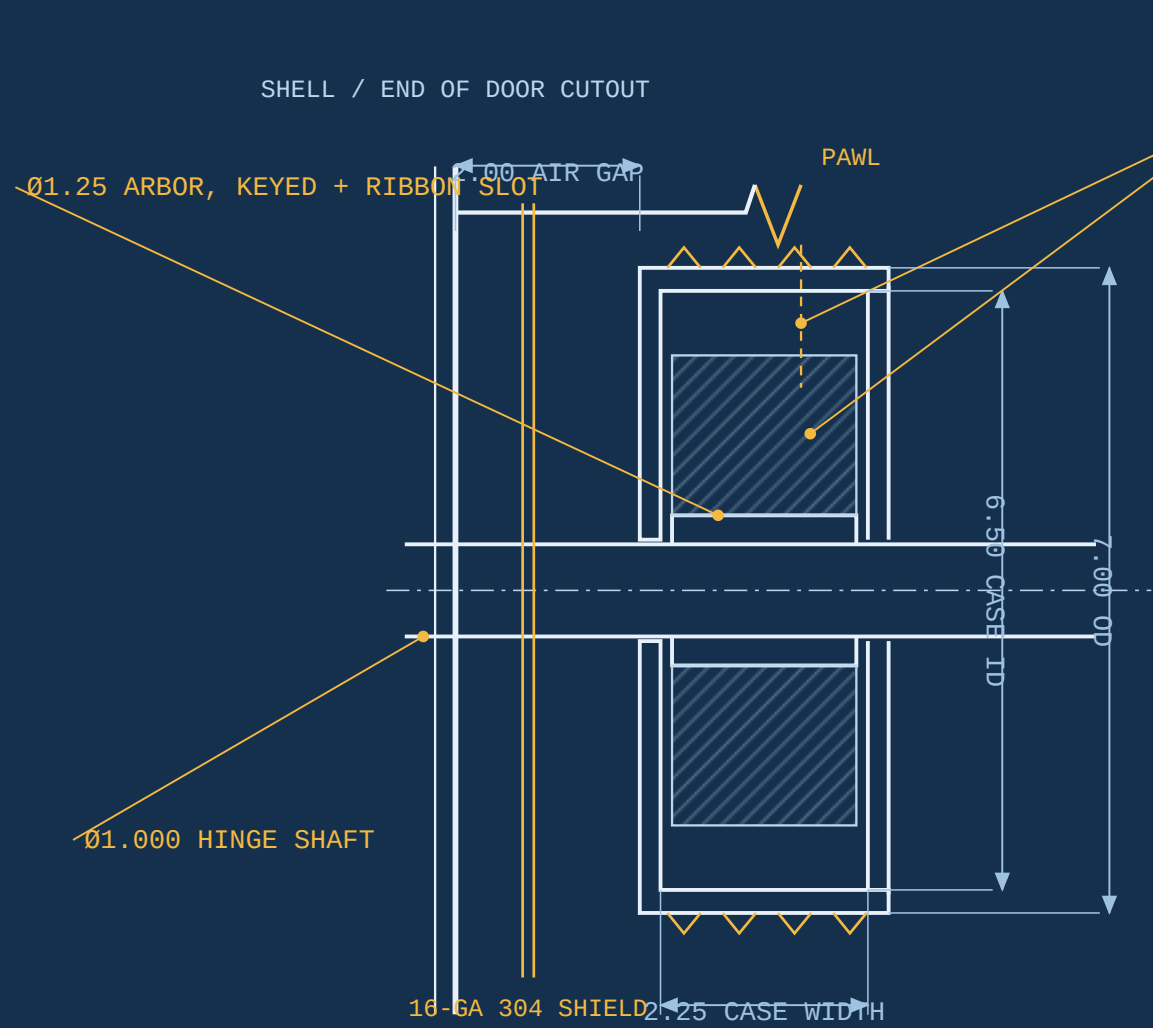
OFFSET SMOKER — DOOR SPRING-HINGE ASSEMBLY

SNAP CAM & FOLLOWER DETAIL

SHT 2/4 · 55 PX/IN

UNITS: INCHES · REV A — PROPOSAL

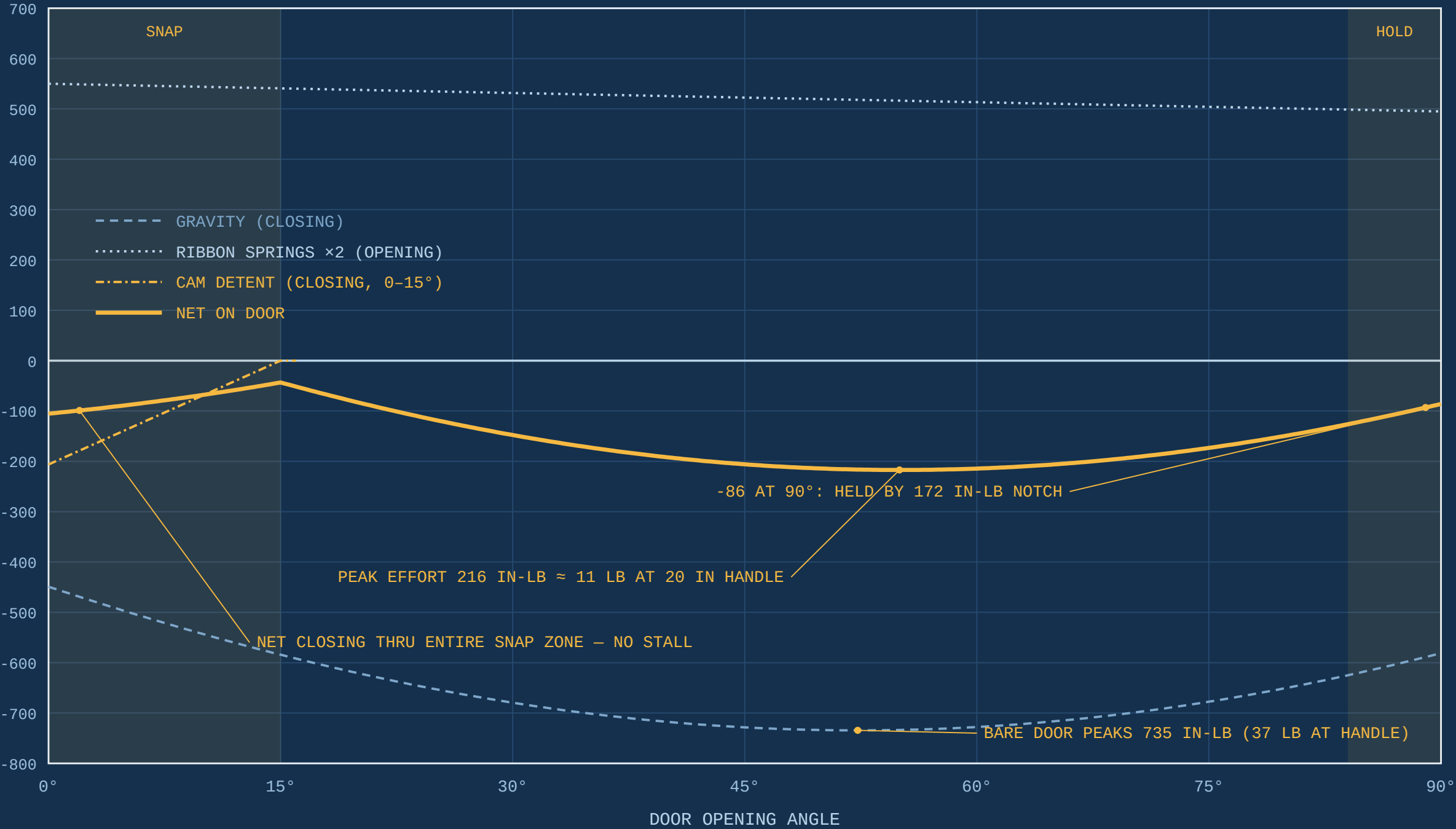
2026-07-04



- NOTES:**
1. SPRING: 2.00 × .085 CR-SI RIBBON, 175 IN LG (14.6 FT), K ≈ 110 IN-LB/TURN. INNER END KEYED TO ARBOR (ON SHAFT), OUTER END HOOKED TO CASE.
 2. NOMINAL PRELOAD 2.5 TURNS/SPRING ($\sigma = 114$ KSI). MAX 3.25 TURNS ($\sigma = 150$ KSI) – STAMP LIMIT ON CASE.
 3. CASE ROTATES IN BRACKET COLLAR FOR PRELOAD; PAWL HOLDS.
 4. ⚠ STORED ENERGY ~250 J WOUND. INSTALL KEEPER PIN BEFORE REMOVING ANY BRACKET FASTENER.
 5. SOURCING: CUSTOM POWER SPRING (VULCAN, JOHN EVANS' SONS). BUDGET ALT: GARAGE-DOOR TORSION STOCK, SAME SHAFT + CAM – SEE DESIGN.MD §7.

OFFSET SMOKER – DOOR SPRING-HINGE ASSEMBLY	
POWER-SPRING CANISTER – SECTION & FACE	SHT 3/4 · 48 PX/IN
UNITS: INCHES · REV A – PROPOSAL	2026-07-04

TORQUE ABOUT HINGE, IN-LB (OPENING +)



OFFSET SMOKER – DOOR SPRING-HINGE ASSEMBLY

TORQUE BALANCE VS DOOR ANGLE

SHT 4/4 · CHART

UNITS: INCHES · REV A – PROPOSAL

2026-07-04